Articulate3D Challenge — Technical Report

2nd Place on Both Test Leaderboards

Frozen USDNet, Geometric Priors, and Structural Post-Processing

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Abstract

Articulate3D evaluates scene-level movable-part articulation and interaction-element segmentation [1]. Starting from released USDNet checkpoints, we combine geometric refinement and structural mask splitting with later full-network flip finetuning and ensembling. Final test scores are 0.4056 on the movable ranking metric and 0.3286 AP_{50} on interaction, placing second on both test leaderboards. Validation is in-sample. Test ground-truth annotations were not accessed directly, but server scores and unlabeled test predictions informed development.

Team

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Final standing (boards frozen 2026-08-28): **1st** on Movable-dev ($AP_{50}^{\text{axis+origin}} = 0.5038$) and Interaction-dev ($AP_{50} = 0.6446$); **2nd** on Movable-test (0.4056) and Interaction-test (0.3286). The extended report, "Geometric Refinement and Structural Post-Processing for Scene-Level Articulated Part Understanding," is publicly available in the code repository.

Method

Movable-part matches require mask IoU above 0.5, sign-invariant axis error below 15°, and (for rotations) origin-to-axis-line distance below 0.25 m. The interaction metric is mask AP_{50}. Baseline errors show axis regression collapsing toward gravity and origins shrinking toward part centroids.

1. **Initial frozen segmentation.** Minkowski U-Net backbone [2], Mask3D decoder [3], mask head and class head are kept at the released weights, *including BatchNorm running statistics*. AP_{50} is therefore bit-identical to the baseline within a checkpoint for articulation-only edits. Structural splitting and later full-network finetuning explicitly change the mask predictions.
2. **One trained component.** An 88 K-parameter attention-pooled origin head (scoring MLP plus origin MLP) is finetuned with the rest of the network frozen. The comparison measures this changed head under a frozen-backbone setup; it does not isolate attention aggregation alone. Two implementation issues affected early runs: (a) BatchNorm buffers can drift during training and confound validation comparisons even when parameters are frozen; (b) resuming a released checkpoint restarts USDNet's coarse-to-fine mask curriculum, which dilates the GT masks by 0.1 m for 100 epochs and caused severe training degradation unless disabled.
3. **Training-free geometric refinement.** Translation axes use the predicted mask's PCA panel normal, or an in-plane direction when a sliding partner leaf exists; rotation axes are forced vertical. Rotation origins are snapped to a mask endpoint perpendicular to the axis. If a predicted handle is near the part, the endpoint farther from it is selected; otherwise the endpoint nearest the model's predicted origin is used. Handles come from the companion interactable-part model. The initial head-plus-geometry recipe takes test $AP_{50}^{\text{axis+origin}}$ from 0.264 to 0.3385 with the segmentation branch untouched.

4. **Structural post-processing (both tasks).** The model routinely merges the two leaves of a double door into one instance. Offline DBSCAN [4] connected-component splitting + fragment filter + NMS, tuned with the official evaluator on validation, improves test scores (movable 0.3385 → 0.3760; interaction 0.2686 → 0.2886).
5. **Full-network finetuning and ensemble.** Later train+validation finetunes update the full network with flip augmentation and the curriculum disabled. Interaction: the `v5`+`v6` union reaches 0.3240; adding `v5b` gives 0.3286. Movable: a single `mov_v11` checkpoint with geometric and structural refinement gives 0.4056. Validation is in-sample; these cumulative competition gains are not controlled multi-seed ablations.

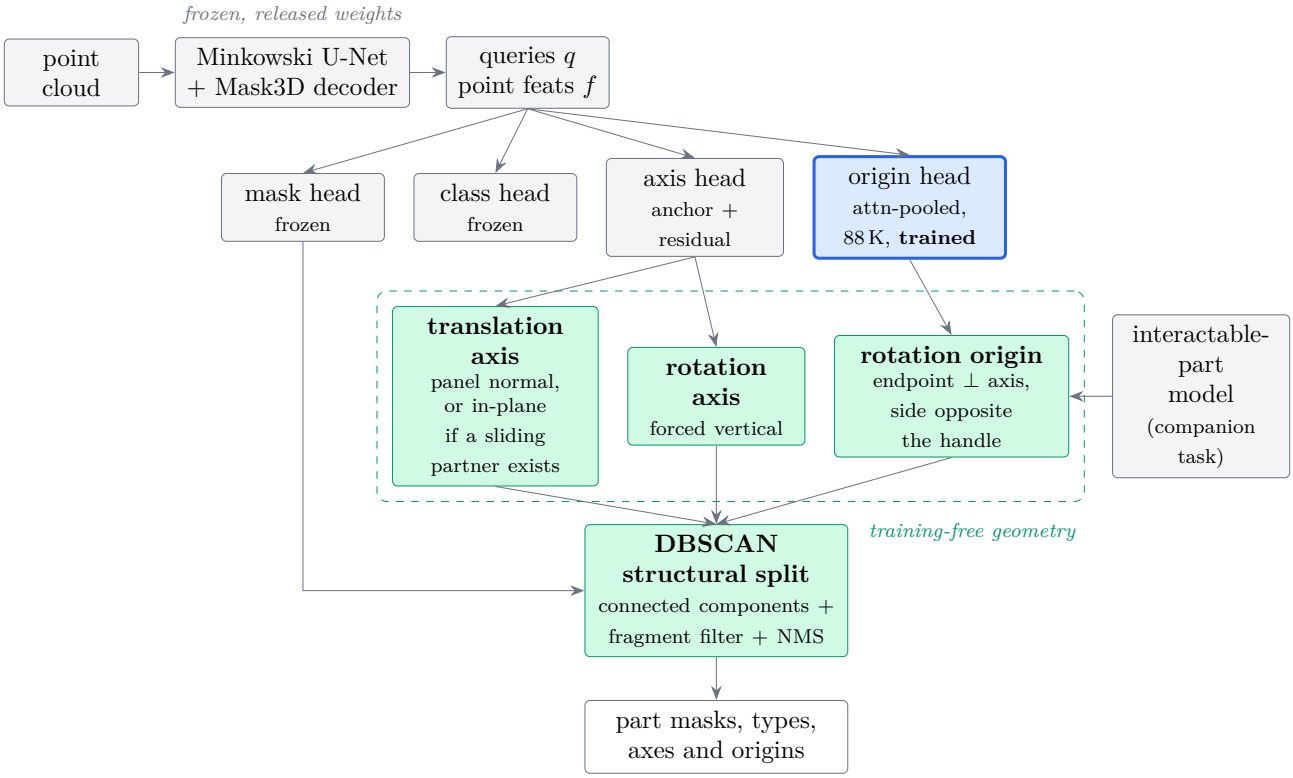


Figure 1: **Method overview.** Initial frozen-model stage only. Grey: frozen at released weights. Blue: the trained origin component. Green: training-free geometry read off the predicted masks and the offline structural post-processing.

Results (challenge server, ranking metric)

Configuration (cumulative)	Movable test $AP_{50}^{axis+origin}$	Interaction test AP_{50}
USDNet baseline (reproduced)	0.2640	0.2686
+ trained origin head + geometric priors	0.3385	—
+ DBSCAN structural post-processing	0.3760	0.2886
+ flip finetune / ensemble (final)	**0.4056**	**0.3286**
Relative to baseline	+54%	+22%

Validation is in-sample for the released checkpoints and later finetunes. Test ground-truth annotations were not accessed directly; repeated server scores and review of unlabeled test predictions informed development, including the sliding-panel height filter.

References

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[3] J. Schult, F. Engelmann, A. Hermans, O. Litany, S. Tang, and B. Leibe. Mask3D: Mask Transformer for 3D Semantic Instance Segmentation. In *Proceedings of the IEEE International Conference on Robotics and Automation*, 2023.

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